



Robot and Smart Card Operating System

Secure, Card-Authenticated Robotic Control powered by a Lightweight Smart Card Operating System



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Introduction

Robots and smart cards are usually separate technologies. This project combines them: before the robot moves, it checks the user's identity and permissions using a Smart Card Operating System (SCOS).

Instead of a password, the robot reads a smart card. The card's own OS checks the user, then tells the robot exactly what it is allowed to do.

7816

Standard for contact smart cards

<200ms

Goal time to check the card

3

Access levels: Guest /
Operator / Admin

Problem Statement & Motivation

THE GAP

- 1 Cheap robots often accept commands from anyone nearby — there is no identity check.
- 2 Simple passwords on small controllers are weak and easy to share or guess.
- 3 Robots used in factories or campuses need to know exactly who gave each command.

THE OPPORTUNITY

- 1 Smart cards already give safe, hard-to-copy identity in banking and transit systems.
- 2 A Smart Card OS can store keys and passwords far more safely than the robot's own memory.
- 3 Combining the two gives a cheap, swappable, secure way to control a robot.

Objectives

1

DESIGN

Build a mobile robot that only obeys commands after identity is checked.

2

SECURE

Set up a Smart Card OS that stores keys, PINs and user roles.

3

INTEGRATE

Connect the card and the robot so commands are passed correctly.

4

VALIDATE

Test how fast and how correctly the whole system works.

Understanding Robots

A robot is a programmable machine that can sense its surroundings, process information, and act — on its own or by remote command.



Mobile Robots

Robots on wheels that move and deliver things — used in this project



Industrial Arms

Fixed robotic arms used for assembly and welding



Service Robots

Help out in homes, hospitals, and stores



Autonomous Vehicles

Self-driving cars and drones used for logistics

Understanding Smart Cards

A smart card is a pocket-sized card with a tiny processor and secure memory. Unlike a magnetic-stripe card, it can compute, store keys, and run a small operating system.



Contact Cards

ISO/IEC 7816

- Has a gold contact pad
- Inserted into a reader
- Used in SIM cards and bank cards
- Reliable, direct connection



Contactless Cards

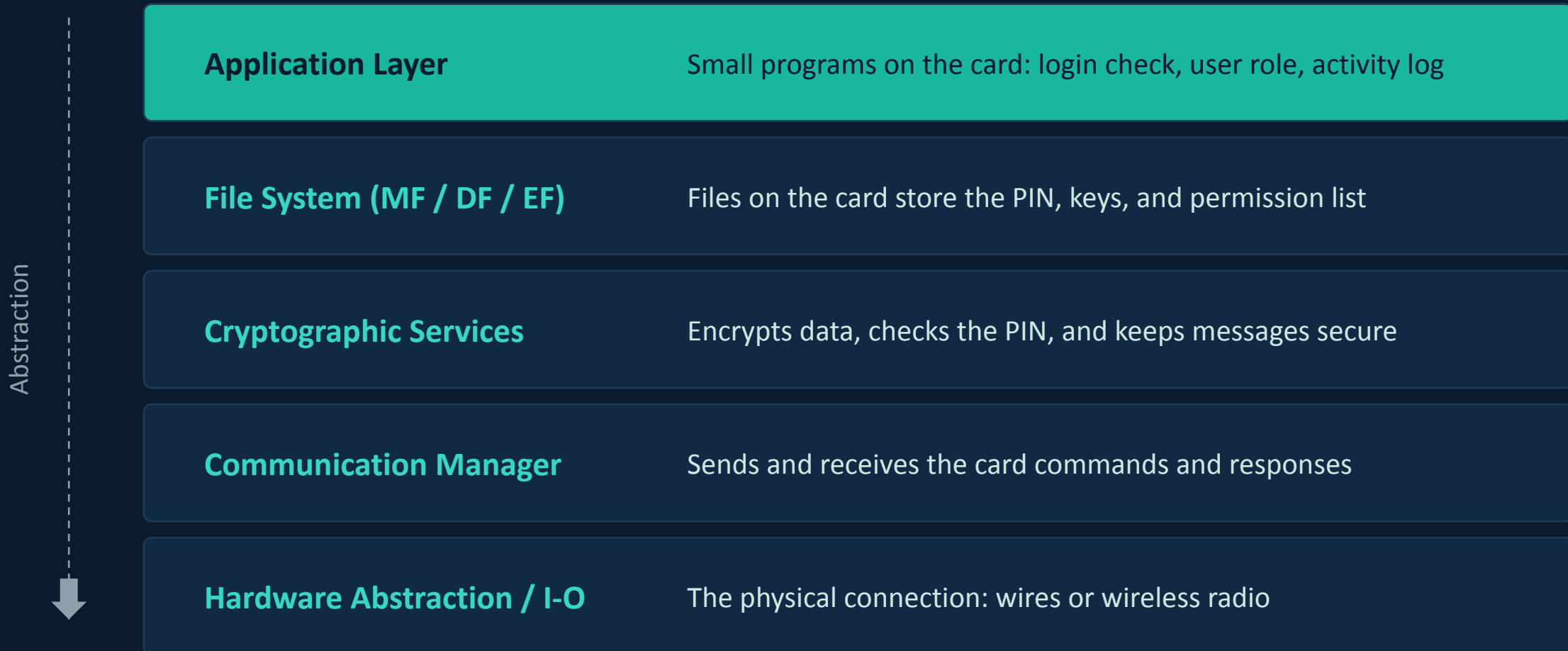
ISO/IEC 14443

- Uses RFID / NFC wireless signals
- Tap and go, no insertion needed
- Used for transit cards and access badges
- Faster to use

CORE CONCEPT

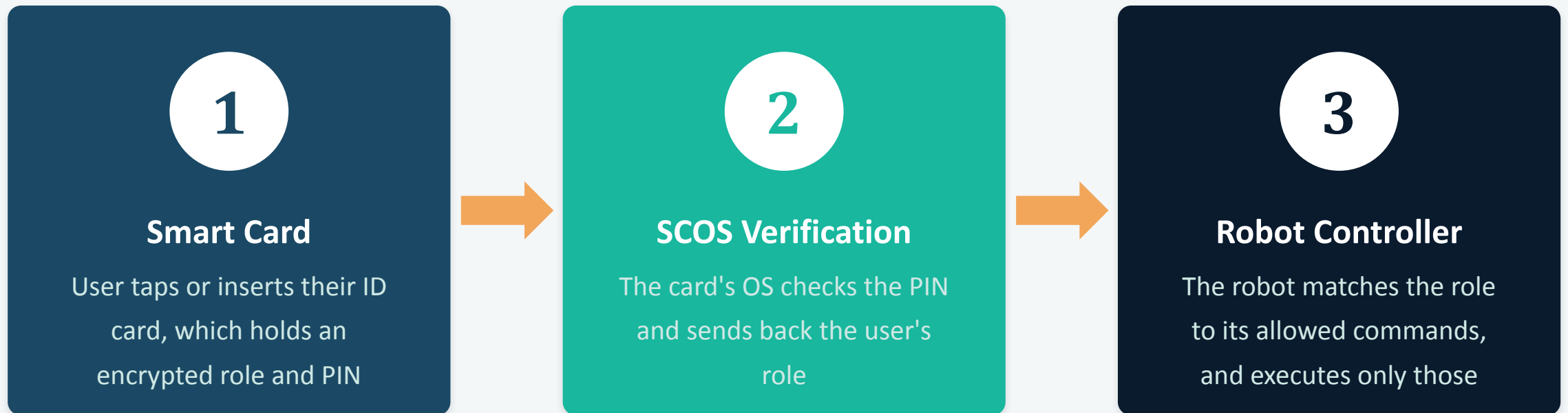
Smart Card Operating System (SCOS)

A small, secure operating system that lives inside the card's chip. Other devices talk to it using simple commands called APDUs.

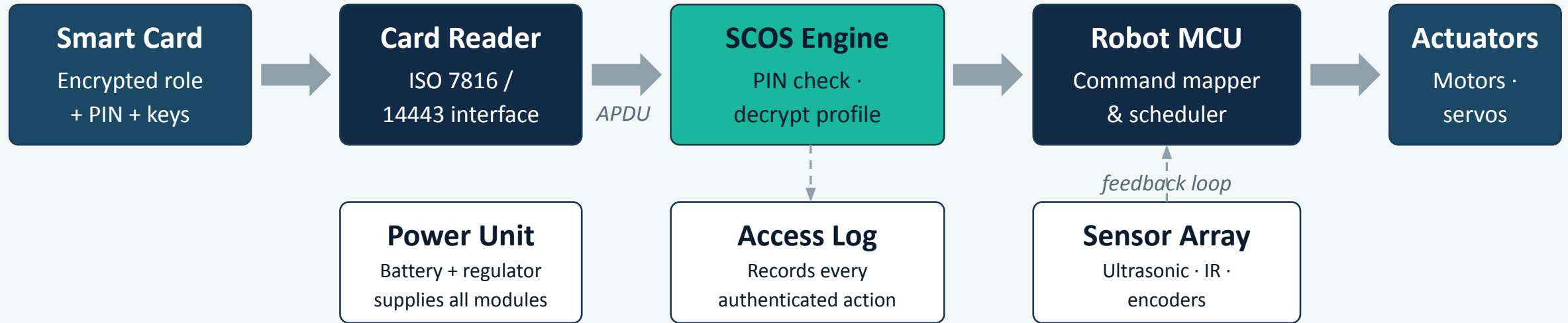


Proposed System Concept

We propose a robot that only runs commands after a smart card proves its identity to the SCOS. The card doesn't just unlock the robot — it also carries the user's exact permissions.



System Architecture



Optional: Wi-Fi or Bluetooth link to a monitoring dashboard

Key idea: the motors never get a raw command directly. Every instruction is first checked against the role and permissions read from the smart card.

Hardware Components



Microcontroller

A small chip (ATmega328 / STM32) that runs the robot's control code



Smart Card Module

Reads both contact and contactless (NFC) smart cards



Motor Driver

Circuit that drives the two wheel motors



Chassis & Wheels

Two-wheel drive chassis with a support caster wheel



Sensor Suite

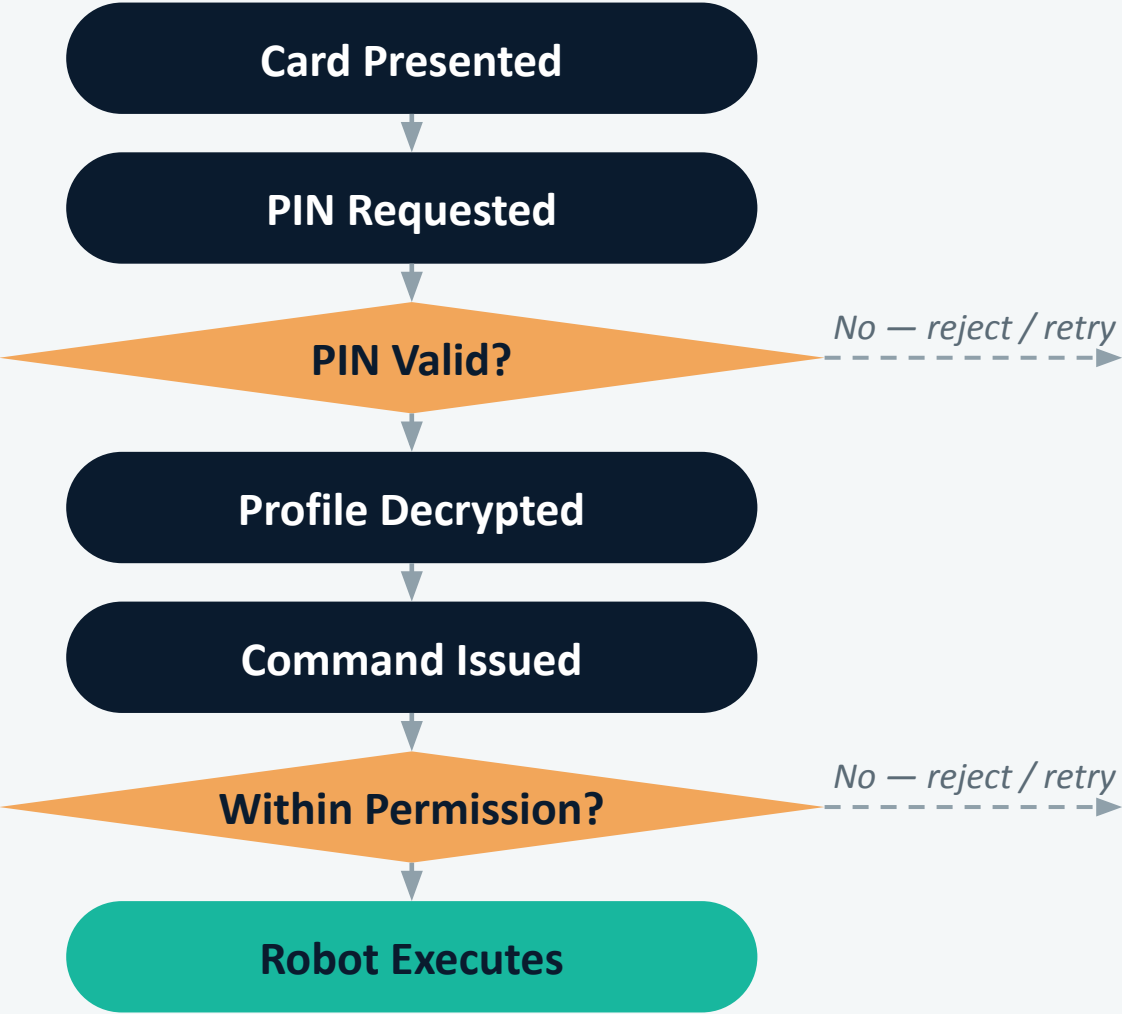
Distance, line-following, and wheel-speed sensors



Power Module

Rechargeable battery pack with a voltage regulator

Working Methodology



Card Presented

User taps or inserts the smart card into the reader.

PIN Requested

The system asks for a PIN using a keypad or app.

Profile Decrypted

If correct, the user's role and allowed commands are sent to the robot.

Robot Executes

The robot runs a command only if it is allowed for that role.

Command Execution & Access Control

Every command sent to the robot is checked against the role read from the card. The table below shows what each role can do.

Command	Guest	Operator	Admin
Move Forward / Reverse			
Turn Left / Right			
Adjust Speed Profile			
Access Sensor Data Log			
Reprogram Task Sequence			
Update Card Permissions			
Firmware / OTA Update			

Applications

1

Warehouse & Logistics

Only trained operators can send out delivery robots

2

Hospitals

Nurses tap their ID card to run medicine-delivery robots

3

Campus & Office Security

Patrol robots run only for verified security staff

4

Manufacturing Floors

Robotic arms can't be reprogrammed by unauthorized staff

5

Smart Homes

Family members get different control levels over home robots

6

Defense & Restricted Zones

Card + PIN required before sensitive robot actions

Advantages



Two-factor security: a smart card is much harder to copy than a password is to guess.



Every action can be traced back to the card holder.



Guests, operators, and admins each have clearly separate permissions.



The same card can be reused across many robots.



Security logic lives on the card, so the robot's own code stays simple.



Built on existing ISO card standards, so it works with existing systems.

Challenges & Limitations



Latency

Checking the card adds a small delay each time — must be planned for.



Environmental Exposure

Outdoor robots need tougher, dust- and vibration-resistant readers.



Card Loss / Theft

A stolen card without PIN protection could still give basic access.



Cost per Unit

A smart card and reader cost more than a simple keypad.



Limited On-Card Memory

The card's small memory limits how much permission data it can store.



Firmware Update Complexity

Updating the card's software needs a secure process.

Traditional vs Smart-Card-Enabled Robots

Criteria	Traditional Robot	Smart-Card-Enabled Robot
Access Control	Open pairing or shared password	Per-user card + PIN, role-scoped
Auditability	Little to no action trace	Every command logged to a card ID
Credential Revocation	Requires firmware/password change	Instantly disable a single card
Onboarding New Users	Manual reconfiguration	Issue a pre-provisioned card
Tamper Resistance	Software-only, easily bypassed	Hardware secure element on card
Hardware Cost Impact	Lower	Slightly higher (reader + card)

Future Scope



Biometric Fusion

Add fingerprint or face check along with the card



Cloud Fleet Management

Track all robots' access logs from one dashboard



Dynamic Role Updates

Update a card's permissions remotely and securely



AI-Assisted Anomaly Detection

Detect unusual command patterns that may mean misuse



Multi-Robot Card Federation

Let one card safely operate a whole fleet of robots



Miniaturized Secure Element

Move the SCOS into a wearable device for hands-free login

Conclusion

This project shows that combining a mobile robot with a Smart Card Operating System is a practical, low-cost way to add strong identity and access control to robotics.

- 1 SCOS gives each operator a secure, portable identity
- 2 Guest, operator, and admin actions stay clearly separated
- 3 Using ISO standards keeps the system compatible and extendable
- 4 The idea scales from a single classroom robot to a full industrial fleet

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Thank You

Questions & Discussion Welcome

